# 7 Probability Distributions



**Johann Carl Friedrich Gauss**
(April 30, 1777 – Feb. 23, 1855)

## Introduction

The frequency distributions are of two types namely Observed frequency distribution and Theoretical frequency distribution. The distributions which are based on actual data or experimentation are called the Observed Frequency distribution. On the other hand, the distributions based on expectations on the basis of past experience are known as Theoretical Frequency distribution or Probability distribution.

Johann Carl Friedrich Gauss (30 April 1777 – 23 February 1855) was a German mathematician and physicist who made significant contributions to many fields in mathematics and sciences. Gauss had an exceptional influence in many fields of mathematics and science, and is ranked among history's most influential mathematicians

In discrete probability distribution we will discuss Binomial and Poisson distribution and the Normal Distribution is a continuous probability distribution

### Learning Objectives

After studying this chapter students are able to understand

- Concept of Bernoulli trial
- Binomial , poisson and normal density function
- Mean and variance of binomial and poisson distribution
- Properties of normal probability curve

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## 7.1 Distribution

The following are the two types of Theoretical distributions :

1. Discrete distribution
2. Continous distribution

### Discrete distribution

The binomial and Poisson distributions are the most useful theoretical distributions for discrete variables.



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